

Unit 11, Lesson 8: Putting it All Together

Benchmarks

MA.7.AR.4.1 Determine whether two quantities have a proportional relationship by examining a table, graph or written description.

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation or a written description.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table or equation.

Learning Targets

- ☐ I can determine if two quantities have a proportional relationship given a real-world context.
- ☐ I can graph a proportional relationship from a table, an equation, or a written description.
- ☐ I can determine the constant of proportionality of a proportional relationship.
- ☐ I can translate a given representation of a proportional relationship to a written description, table, or equation.

Some people say that taking a bath uses more water than a shower. Others disagree.

- ©Accelerate Learning Inc. – All Rights Reserved

11.8.2 Activity: Station 1 – Gym Membership Plans

In January, Gigi signed up for a membership at Total Fitness. The plan she chose costs \$65 in start-up fees and then \$20 per month starting in February. Julien also signed up at Total Fitness in January. His plan costs \$32 per month starting in February, and his start-up fees and January costs were waived.

1. Complete the tables to represent the total costs of their gym memberships for both Gigi and Julien for the first 8 months of their memberships.

Gigi’s Membership Costs	
Months Since January	Total Cost
0	
1	
2	
3	
4	
5	
6	
7	

Julien’s Membership Costs	
Months Since January	Total Cost
0	
1	
2	
3	
4	
5	
6	
7	

2. Determine whether either or both gym memberships are a proportional relationship. Then, complete the statements.

Gigi’s membership costs

☐ represent

☐ do not represent

a

proportional relationship because _____

_____.

Julien's membership costs

- ☐ represent
- ☐ do not represent

a

proportional relationship because _____

_____.

3. For the relationship(s) that is proportional, write an equation to represent the total cost, c , in dollars, for m months of the membership.
4. Discuss with your partner which membership seems to be a better deal. Then, summarize your discussion.

11.8.3 Activity: Station 2 – Who Earns More Money?

Tia works at the Boys and Girls Club after-school program at Lockhart Middle School in Orlando, FL. The table shows how much money she earned each day she worked last week.

Time Worked (hours)	1.75	2.5	4.25
Money Earned (\$)	18.55	26.50	45.05

Kahlil has a job bagging groceries at Publix where he earns \$9.75 per hour.

1. Who would make more money if they both work 10 hours? Explain or show work.

2. On the coordinate grid shown, draw a graph that represents y , the amount of money, in dollars, Tia makes for working x hours. Label both axes appropriately.



3. On the same grid, draw a graph that represents y , the amount of money, in dollars, Khalil makes for working x hours.
4. Label each graph with the name of the person whose earnings it represents.

5. Explain how the graph shows which person has the greatest hourly rate of pay.
6. For each person, determine the constant of proportionality and write an equation to represent the relationship between their total earnings, y , in dollars, and the hours worked, x .

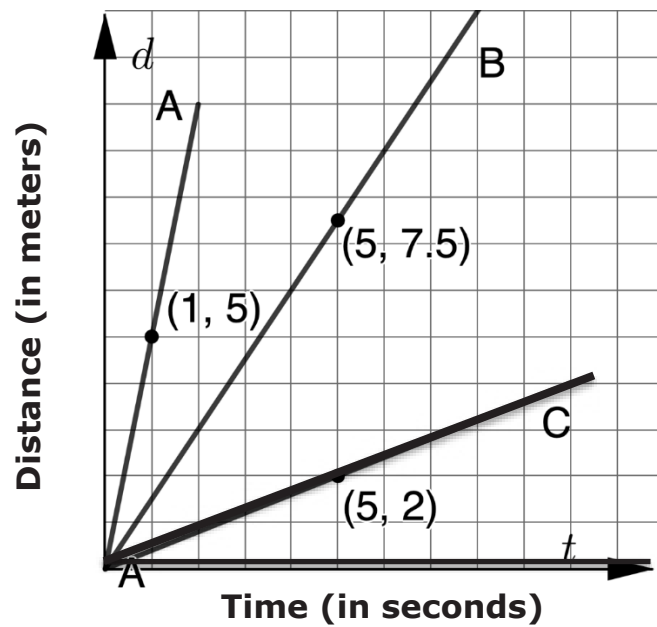
Tia’s Earnings (\$)	Constant of Proportionality	
	Equation	$y = \underline{\hspace{2cm}}$

Khalil’s Earnings (\$)	Constant of Proportionality	
	Equation	$y = \underline{\hspace{2cm}}$

11.8.4 Activity: Station 3 – Robot Races

The students in the STEM Club at Umatilla Middle School built solar-powered robots which they raced in the gym at their STEM Showcase.

The graphs shown represent the distance d , in meters, that each of the three robots A, B, and C traveled after t seconds.



1. Determine whether each statement in the table is true or false.

Statement	True or False?
Robot B traveled in a different direction than the other robots.	☐ true ☐ false
Robot B traveled 5 meters in 7.5 seconds.	☐ true ☐ false
Robot B traveled 7.5 meters in 5 seconds.	☐ true ☐ false
Robot B traveled $\frac{2}{3}$ meters per second.	☐ true ☐ false
Robot B traveled $\frac{3}{2}$ meters per second.	☐ true ☐ false

2. Determine the constant of proportionality, $\frac{d}{t}$, for each robot.

Robot	Constant of Proportionality
A	
B	
C	

3. Explain what the constant of proportionality determined for each robot represents in this context.
4. Write an equation to represent the relationship between distance, d , and time, t , for each robot.

Robot	Equation
A	
B	
C	

5. Assume the robots traveled at the same rate for 10 seconds. How far would each robot have traveled in that time?

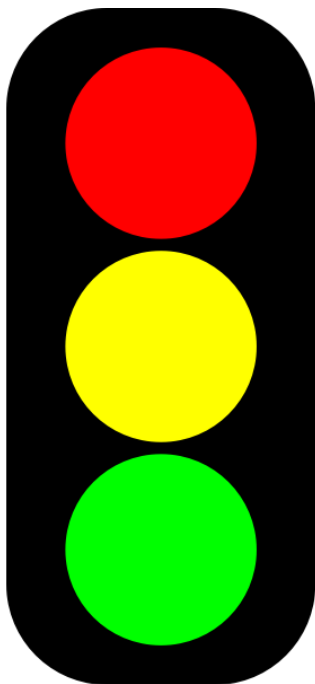
Robot	Distance Traveled in 10 seconds (meters)
A	
B	
C	

11.8.5 Wrap-Up: Reflection

So far in this unit, you have learned multiple concepts about proportional relationships including those listed below.

- Determining if two quantities have a proportional relationship
- Determining the constant of proportionality of a proportional relationship given a table, graph, or written description
- Graphing proportional relationships from a table, equation, or written description
- Translating between different representations of proportional relationships

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck about how to...

I'm starting to get the hang of...

I feel confident and ready to go with...

